



Since 1983

Dr. Vishwanath Karad

**MIT WORLD PEACE
UNIVERSITY** | PUNE

TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS



M.Sc. Biotechnology

Programme Structure

 mitwpu.edu.in

Division	Faculty of Science & Health Science
School Name	School of Science & Environmental Studies
Department Name	Department of Biosciences & Technology
Programme Name	M.Sc. Biotechnology

+ + + + + + + + + + + + + + + **CREDIT DISTRIBUTION** + + + + + + + + + + + + + + +

| Course Type | Description |
|----------------------|--|
| Programme Core | Courses dealing with foundations, depth and breadth of the major in which a student is admitted at MIT-WPU |
| Programme Electives | Open electives under the Programme allow students to specialise in a particular area connected to their major. |
| University Core | Courses that reflect the core MITWPU values and the mission of Life Transformation of students. |
| University Electives | Multidisciplinary courses across the faculties at MIT-WPU and outside the Programme core. |

M.Sc. (Biotechnology) (First Year) (2025-27)
Semester – I

| Sr. No | Name of the Course | Type | Credits |
|--------|--|------|-----------|
| 1 | Biochemistry | PC | 2 |
| 2 | Microbiology | PC | 2 |
| 3 | Molecular and Cell Biology | PC | 2 |
| 4 | Biostatistics | PC | 3 |
| 5 | Biotechnology Lab – 01 | PC | 3 |
| 6 | Research Methodology | PC | 4 |
| 7 | Program Elective-01 | PE | 4 |
| 8 | Scientific Studies of Mind, Matter, Spirit and Consciousness | UC | 2 |
| 9 | Yoga | UC | 1 |
| | Total | | 23 |

M.Sc. (Biotechnology) (First Year) (2025-27)
Semester – II

| Sr. No | Name of the Course | Type | Credits |
|--------|------------------------------------|------|-----------|
| 1 | Bioanalytical Techniques | PC | 3 |
| 2 | Bioprocess Technology | PC | 3 |
| 3 | Biotechnology Lab – 02 | PC | 3 |
| 4 | Genomics & Genetic Engineering | PC | 3 |
| 6 | Immunology | PC | 3 |
| 7 | Program Elective-02 | PE | 4 |
| 8 | Peace Building: Global Initiatives | UC | 2 |
| | Total | | 21 |
| | | | |

M.Sc. (Biotechnology) (Second Year) (2025-27)
Semester – III

| Sr. No | Name of the Course | Type | Credits |
|--------|---|------|-----------|
| 1 | Computational Biology & Bioinformatics | PC | 3 |
| 2 | Regenerative Biology & Stem Cell Technology | PC | 3 |
| 3 | Systems and Synthetic Biology | PC | 3 |
| 4 | Program Elective-03 | PE | 4 |
| 5 | On Job Training (OJT)/Internship | PC | 4 |
| 6 | Project | PC | 5 |
| | Total | | 22 |

M.Sc. (Biotechnology) (Second Year) (2025-27)
Semester – IV

| Sr. No | Name of the Course | Type | Credits |
|--------|---------------------------------------|------|-----------|
| 1 | Bio-entrepreneurship, IPR & Bioethics | PC | 3 |
| 2 | Biomimetics & Nanobiotechnology | PC | 3 |
| 3 | Scientific Communication | PC | 2 |
| 4 | Program Elective-04 | PE | 4 |
| 5 | Project/Industrial Internship | PC | 10 |
| | Total | | 22 |

M.Sc. (Biotechnology) (Second Year) (2025-27)
Semester – IV

| Sr. No | Name of the Course | Type | Credits |
|--------|---------------------------------------|------|-----------|
| 1 | Bio-entrepreneurship, IPR & Bioethics | PC | 3 |
| 2 | Biomimetics & Nanobiotechnology | PC | 3 |
| 3 | Scientific Communication | PC | 2 |
| 4 | Program Elective-04 | PE | 4 |
| 5 | Project/Industrial Internship | PC | 10 |
| | Total | | 22 |

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.